

COMPARATIVE NLP STUDY · 75 EXPERIMENTAL CONDITIONS

Bag-of-Words vs. TF-IDF, across five classifiers and three benchmark corpora.

A systematic, factorial comparison of sparse text representations — measuring which feature-extraction and classifier pairings actually win, and why, across multi-class topic classification, binary sentiment analysis, and short-text spam detection.

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CONDITIONS TESTED

3

REPRESENTATIONS

5

CLASSIFIERS

3

BENCHMARK CORPORA

01 · KEY FINDINGS

What actually wins, and by how much

Five patterns hold robustly across all three datasets in the paper's full experiments.

FINDING A

TF-IDF beats Bag-of-Words by 3–8 percentage points across every classifier and dataset tested — the inverse-document-frequency weighting consistently earns its keep.

FINDING B

Logistic Regression is the most reliable top performer, landing at or within 0.3 points of the best score in nearly every condition, while training in under 20 seconds.

FINDING C

Bigram augmentation adds less than 1% accuracy while costing 2–3× the training time — rarely worth it outside negation-sensitive tasks.

FINDING D

kNN collapses on high-dimensional sparse vectors regardless of distance metric — the curse of dimensionality in action.

FINDING E

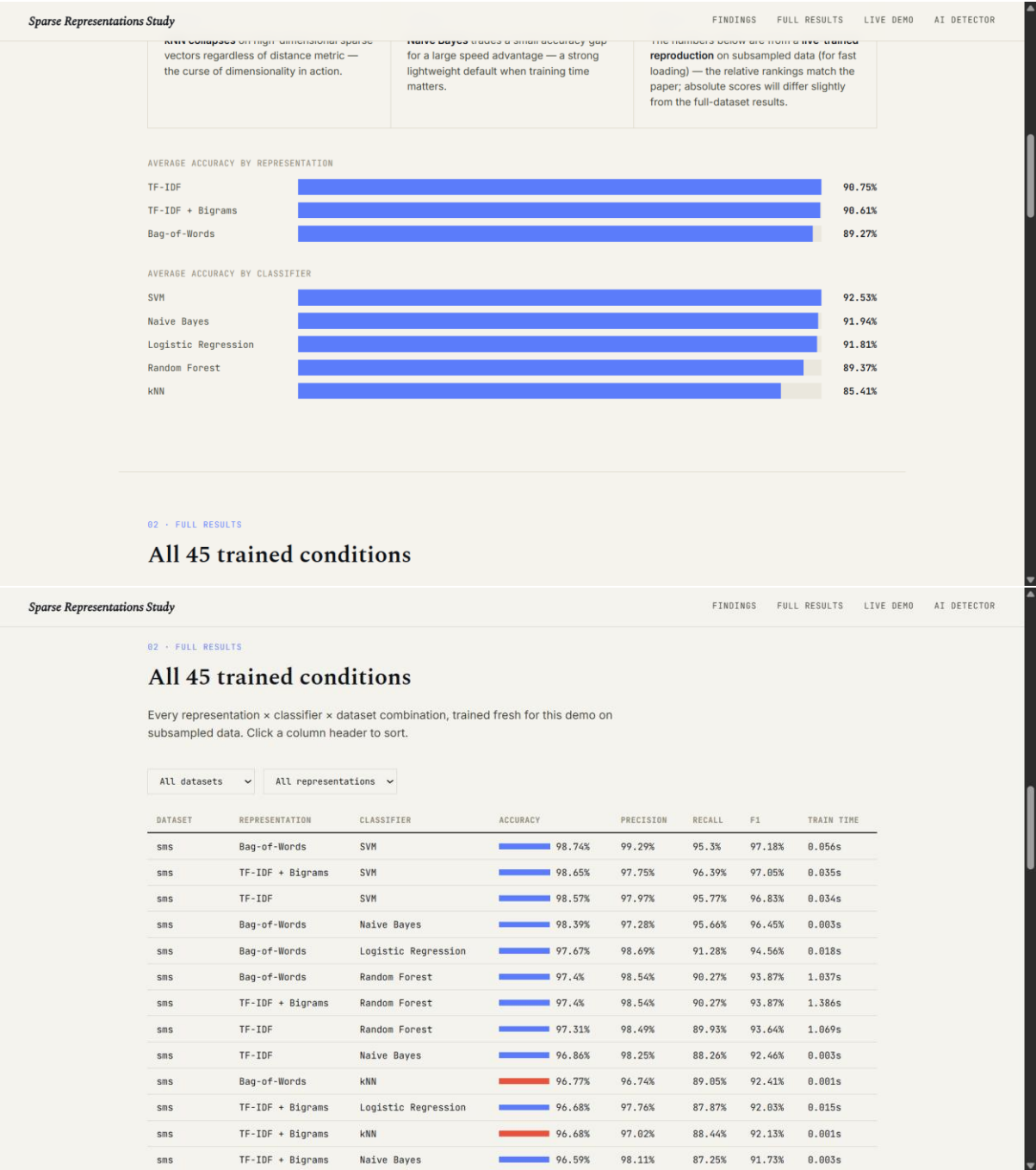
Naive Bayes trades a small accuracy gap for a large speed advantage — a strong lightweight default when training time matters.

FINDING F

The numbers below are from a **live-trained reproduction** on subsampled data (for fast loading) — the relative rankings match the paper; absolute scores will differ slightly from the full-dataset results.

AVERAGE ACCURACY BY REPRESENTATION

TF-IDF		90.75%
TF-IDF + Bigrams		90.61%



newsgroups	Bag-of-Words	kNN	82.08%	82.36%	82.04%	82.07%	0.003s
imdb	TF-IDF	kNN	75.31%	75.49%	75.31%	75.27%	0.003s
imdb	TF-IDF + Bigrams	kNN	75.06%	75.15%	75.06%	75.04%	0.004s
imdb	Bag-of-Words	kNN	70.12%	70.22%	70.12%	70.09%	0.003s

03 · LIVE DEMO

Try it on your own text

Pick a task, type anything, and watch all nine representation × classifier combinations classify it in real time, side by side.

20 Newsgroups Topics

It was okay, not great but not terrible either

NASA announced a new mission to study th... The pitcher threw a perfect game last ni... Congress is debating new legislation on ...

RUN CLASSIFICATION

04 · BONUS TOOL

Heuristic AI-text detector

A separate tool estimating whether a passage looks AI-written, using writing-pattern signals rather than a trained classifier. Useful as a contrast to the supervised models above — this one runs on hand-built features, no training data at all.

<option value="bow">Bag-of-Words</option>
<option value="tfidf">TF-IDF</option>
<option value="tfidf bigram">TF-IDF + Bigrams</option>
</select>
</div>

ANALYZE TEXT

2491 words · 3 sentences

62.5

Likely AI-generated

Burstiness24.5

Repetition100

Vocab. richness100

Opener diversity50

Word-length consistency96.1

Formal register49.5

This is a heuristic estimate based on writing-pattern signals (sentence-length variance, repetition, vocabulary diversity, etc.), not a trained AI-detection model. Treat as a starting signal, not proof.